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SOFTWARE REQUIREMENTS SPECIFICATION (SRS) For IRO (Issues Reporting Outlet)

A Web-Based Issue Reporting System for AAUA Students and Host Communities

1. Introduction

1.1 Purpose

The Software Requirements Specification (SRS) defines the functional and non-functional requirements for **the Issues Reporting Outlet (IRO)**, a web-based platform designed to facilitate efficient reporting, tracking, and management of issues affecting students of Adekunle Ajasin University, Akungba-Akoko (AAUA), and members of its host communities, including Akungba, Iwaro, Etioro, and Ayegunle.

The primary purpose of the system is to replace inefficient manual reporting methods with a centralized digital platform that enables users to submit complaints, monitor their progress, and receive timely updates while allowing administrators to manage reports effectively.

1.2 Scope

The IRO system is intended to provide an efficient, transparent, and user-friendly mechanism for reporting community and campus-related issues.

The application supports two categories of users: **Students and Indigenes**, who can register, authenticate, submit reports, monitor the status of submitted complaints, and review their reporting history. An **Administrator** is responsible for reviewing reports, updating their status, managing users, and maintaining the integrity of the platform.

The system is designed as a lightweight web application suitable for academic demonstration while providing a scalable architecture that can support future enhancements such as mobile integration, GIS mapping, and notification services.

1.3 Objectives

The objectives of the proposed system are to:

- Develop a centralized online issue-reporting platform.
- Improve communication between citizens and administrators.
- Eliminate delays associated with manual complaint handling.
- Promote transparency through report tracking.
- Improve response time to reported issues.
- Maintain accurate records of reported incidents.
- Provide administrators with tools for monitoring and managing reports.

1.4 Intended Audience

This document is intended for:

- Software Developers
- Project Supervisor
- System Testers
- System Administrators
- Future Developers
- End Users

2. Overall Description

2.1 Product Perspective

IRO is a standalone web-based application developed using HTML, CSS, and JavaScript. The application provides a modern graphical user interface through which users can submit reports concerning environmental, infrastructural, security, and campus-related issues.

The application acts as a bridge between community members and administrators responsible for resolving reported issues.

2.2 Product Functions

The major functions of the system include:

- User Registration
- User Authentication
- Secure Login
- Issue Submission
- Report Tracking
- Report Management
- User Management
- Status Updates
- Administrative Dashboard
- Logout

2.3 User Characteristics

Student

Students are registered users of the platform who can report issues affecting academic activities and campus facilities.

Capabilities include:

- Registration

- Login
- Report Submission
- View Personal Reports
- Track Report Status
- Logout

Indigene

Indigenes are members of the surrounding communities who use the platform to report community-related issues.

Capabilities include:

- Registration
- Login
- Submit Community Issues
- Monitor Submitted Reports
- Logout

Administrator

Administrators supervise the entire platform.

Responsibilities include:

- Reviewing submitted reports
- Updating report status
- Managing registered users
- Removing invalid reports
- Monitoring overall system activities

3. System Features

3.1 User Registration

Description

The system shall allow new users to create an account before accessing reporting services.

Inputs

- Full Name
- Email Address
- Phone Number
- Password
- User Category

Outputs

- Successful registration confirmation
- Validation errors where applicable

3.2 User Login

The system shall authenticate registered users using their email address and password before granting access to protected pages.

3.3 Issue Reporting

Users shall be able to submit reports by providing:

- Issue Title
- Category
- Description
- Location

- Image Attachment (Optional)

Upon successful submission, the system records the report and assigns an initial status of **Pending**

3.4 Report Tracking

Registered users shall be able to monitor the progress of previously submitted reports.

Possible report statuses include:

- Pending
- In Progress
- Resolved

3.5 Administrative Management

The administrator shall:

- View all reports
- Filter reports
- Update report status
- Delete inappropriate reports
- Manage registered users

4. Functional Requirements

Requirement ID	Description
FR-01	The system shall allow users to register.
FR-02	The system shall authenticate registered users before granting access.
FR-03	The system shall allow users to submit issue reports
FR-04	The system shall validate all mandatory fields before submission
FR-05	The system shall store submitted reports.
FR-06	The system shall display the current status of submitted reports
FR-07	The administrator shall view all reports
FR-08	The administrator shall update report status
FR-09	The administrator shall manage registered users
FR-10	The system shall provide secure logout functionality.

5. Non-Functional Requirements

Performance

- Response time shall not exceed five seconds under normal usage.
- The application shall support multiple users simultaneously.

Security

- User authentication shall be required before accessing protected resources.
- Password information shall be securely managed.
- Administrative functions shall be restricted to authorized users.

Reliability

- The system shall maintain data consistency during operations.
- Reports shall remain available until intentionally removed by an administrator.

Usability

The interface shall:

- Be easy to navigate.
- Support users with minimal computer knowledge.
- Provide meaningful validation messages.
- Maintain consistent page layouts.

Maintainability

The software shall use modular development practices, separating HTML, CSS, and JavaScript to simplify maintenance and future updates.

Compatibility

The system shall operate correctly on:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

6. External Interface Requirements

User Interface

The application consists of the following interfaces:

- Home Page
- Registration Page
- Login Page
- User Dashboard
- Report Issue Page

- My Reports Page
- Administrator Dashboard

The interface follows a responsive design that supports desktops, tablets, and smartphones.

Hardware Interface

Minimum hardware requirements include:

- Intel Core i3 Processor or equivalent
- 4 GB RAM
- 500 MB Available Storage

Software Interface

Development Environment:

- Visual Studio Code
- HTML5
- CSS3
- JavaScript
- GitHub
- Draw.io

Deployment Environment:

- Modern Web Browser

7. System Constraints

The following constraints apply:

- Internet connectivity is required for deployment.

- Users must register before submitting reports.
- Only administrators may modify report statuses.
- The current version focuses on front-end implementation due to project scope.

8. Assumptions and Dependencies

The following assumptions were made during system development:

- Users possess basic computer literacy.
- Internet access is available.
- Administrators actively review incoming reports.
- Users provide truthful and complete information.

9. Future Enhancements

Future versions of the system may include:

- Email notifications
- SMS alerts
- Mobile application
- GPS-based location detection
- Interactive map visualization
- Artificial Intelligence for automatic issue categorization
- Analytics dashboard
- Live chat support
- Report prioritization based on urgency

10. Conclusion

The **IRO (Issues Reporting Outlet)** system provides a practical and efficient solution to the challenges associated with traditional issue reporting within AAUA and its host communities. By offering a centralized web-based platform for submitting, tracking, and managing reports, the system improves accessibility, accountability, and communication between stakeholders. The requirements documented in this SRS provide a comprehensive blueprint for the successful implementation, testing, deployment, and future enhancement of the application.